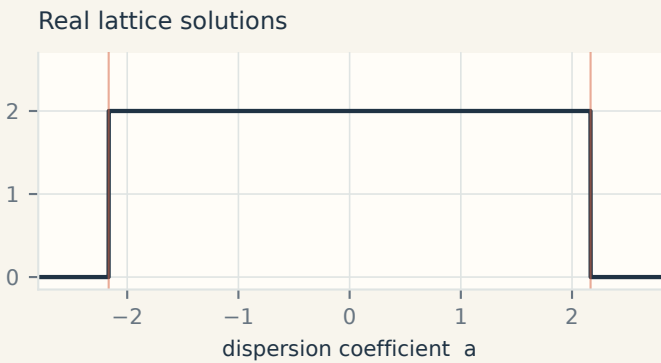
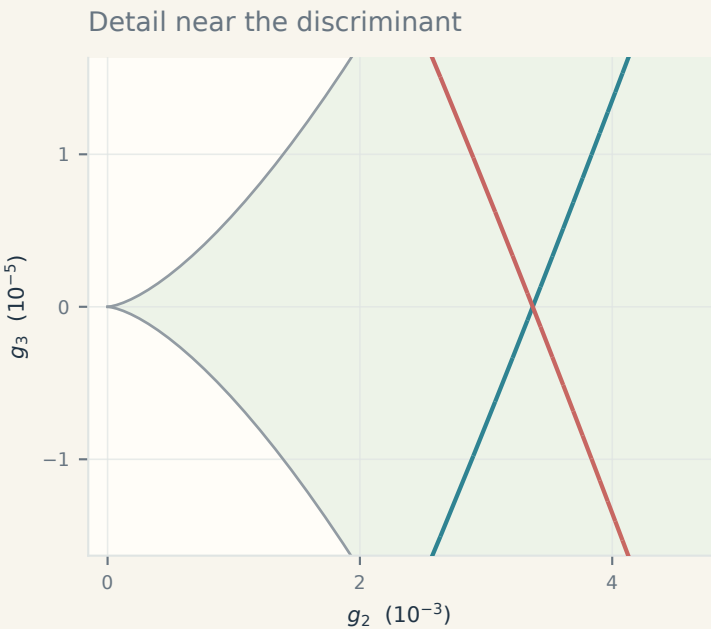
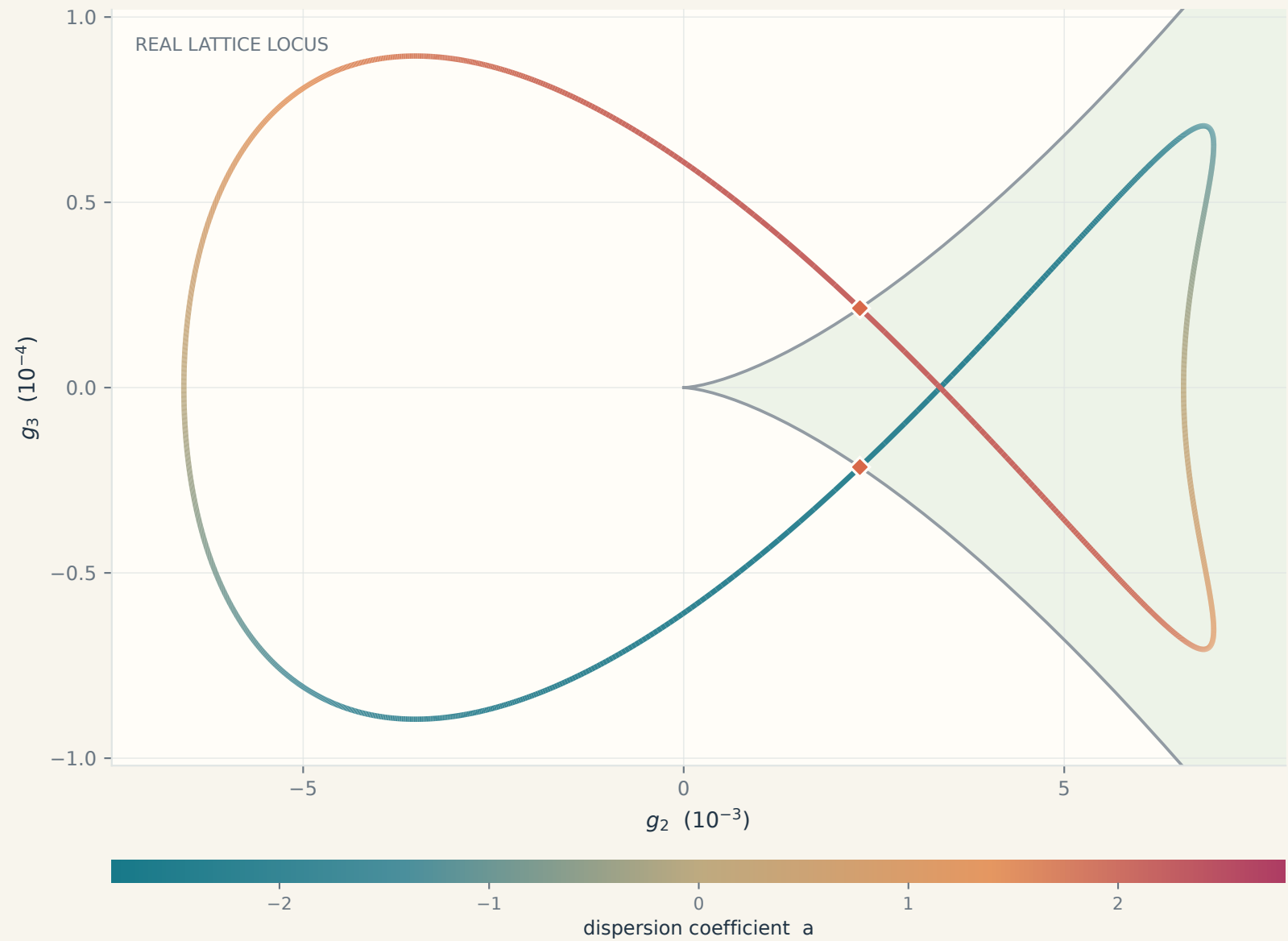


Kawahara · the returning loop

$$u'''' + au'' - u - 840u^2 = 0$$

$$(n, p) = (2, 4) \cdot q = 4$$



Grey cusp: $\Delta = 0$
Pale region: $\Delta > 0$
Diamonds: real degeneration parameters

Two real lattices for $|a| < 13/6$. At each endpoint they merge into one degenerate wave.
Beyond $|a| = 13/6$ the two lattices are complex conjugates and leave this real plane.
Lattice of the full twisted period group Γ · Real invariants do not alone imply a bounded real profile.